

JACOB COOK

Security Engineer | Detection & Response Automation | Threat Modeling · Pen Testing · Python · Cloud · MITRE ATT&CK

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Hands-on security practitioner combining offensive perspective (CompTIA PenTest+, SecurityX) with defensive depth: built a Wazuh SIEM from scratch, authored Sigma rules mapped to MITRE ATT&CK, and engineered Python/FastAPI automation that scales incident response. Strong cloud and identity foundations across Azure and AWS labs. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress. Comfortable applying threat-modeling thinking to detection and incident workflows.

TECHNICAL SKILLS

Detection & Response: Sigma Rule Authoring, MITRE ATT&CK, Threat-Modeling-Driven Detection, Incident Detection & Response, Logging Platform Maintenance, Attack Surface Management Concepts, Threat Hunting

Offensive & Threat Modeling: Pen Testing Concepts (PenTest+ certified), Red-Team Mindset, Threat Modeling, Vulnerability Analysis, OWASP, Phishing & Social Engineering Triage

Automation & Development: Python (FastAPI, REST APIs), Bash, Webhook-Driven Playbooks, SQL, Git, Golang (familiar)

Cloud & Identity: AWS labs, Azure / Entra ID, Okta, SSO / RBAC, Terraform, Linux infrastructure

TECHNICAL PROJECTS

SOAR-lite: Incident Response Orchestrator (Python / FastAPI)

2026

- Developed processes, tooling, and automation to scale incident management response: webhook ingestion, playbook routing, IP blocking, Slack notification, and incident write-up — directly aligned with ClickHouse's "scale IR" focus.
- Eliminated four manual triage steps and reduced triage time by over 60% in lab testing.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against the real FastAPI app, not mocks.

Stryker / Intune Detection Pack: Detection-as-Code

2026

- Applied threat-modeling-driven analysis to derive 6 Sigma-format detection rules mapped to MITRE ATT&CK that eliminated 100% of false positives in a 100-event simulation.
- Documented 18 benign false-positive scenarios as repeatable artifacts that close coverage gaps and improve attack surface awareness.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py).

Blue Team SOC Monitoring Lab: Wazuh SIEM

2026

- Built and maintained a security logging platform (Wazuh) end-to-end — log ingestion, normalization, alert rules — and validated 12 detection patterns against 242 simulated events with 100% precision.
- Reduced false-positive volume by 100% by tuning out 79 benign-only matches without dropping a single true positive.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

Phishing Analysis Lab

2026

- Triaged 5 phishing samples with header inspection, SPF/DKIM/DMARC validation, IOC extraction, and attacker infrastructure mapping — 100% accurate verdicts against ground truth.
- Translated triage outcomes into IOC-driven detection improvements — the same red-team-to-blue-team feedback loop ClickHouse's role centers on.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6)

2024

- Engineered RBAC and Azure AD SSO controls on a customer-facing AI platform — direct cloud security and identity engineering work in production.
- Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response

2019 - 2025

- Maintained 99.5% system uptime by triaging telemetry alerts on rotating shifts — direct experience handling security/operations events at scale.
- Resolved 200+ anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30%.
- Reduced mean time to recovery by an estimated 20% through cross-functional collaboration with engineering and operations teams.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)